

AYUTTHAYA, Thailand

WMO: 484150

Lat: 14.5347N

Lon: 100.7250E

Elev: 32

StdP: 14.68

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	64.6	67.1	51.5	56.7	70.2	55.1	64.7	72.0	23.4	71.7	21.5	71.8	6.1	50	0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	16.9	99.5	80.4	97.8	80.4	96.2	80.2	83.5	93.2	82.7	92.2	82.1	91.4	5.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	161.4	88.8	80.3	157.6	88.0	79.8	154.8	87.3	47.6	94.0	46.7	92.3	46.0	91.4	89.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.2	14.2	11.7	DB	60.7	100.7	4.0	2.7	57.9	102.6	55.5	104.2	53.3	105.7	50.4	107.7	
			WB	54.8	85.3	3.1	1.5	52.5	86.4	50.7	87.2	48.9	88.1	46.7	89.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	84.0	79.8	83.2	85.7	87.5	87.2	85.9	84.8	84.4	83.9	83.9	82.5	79.9
	DBStd	3.65	3.81	2.80	3.31	2.98	2.97	2.16	2.19	1.66	1.86	2.01	2.72	3.72
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12428	923	930	1107	1124	1154	1077	1079	1066	1016	1050	975	927
	CDD65	6953	458	510	642	674	689	627	614	601	566	585	525	462
	CDH74	80860	4377	5698	7971	8848	9017	7819	7366	7021	6413	6532	5458	4340
(13)	CDH80	36844	1808	2630	3999	4795	4733	3701	3220	2903	2502	2582	2221	1750
(14)	Wind	WSAvg	3.4	4.3	3.1	4.0	3.5	3.2	3.2	2.8	1.9	2.5	4.0	5.5
(15) Precipitation	PrecAvg	44.9	0.4	0.3	1.5	3.2	5.7	4.8	5.0	6.1	10.7	6.1	1.3	0.2
	PrecMax	56.6	1.9	1.7	6.9	8.5	10.6	8.1	13.0	10.6	17.3	16.3	6.5	2.2
	PrecMin	33.9	0.0	0.0	0.0	0.3	1.3	0.8	2.0	2.5	5.6	1.5	0.0	0.0
	PrecStd	6.2	0.5	0.4	1.7	2.2	2.5	1.8	2.2	2.1	3.0	2.9	1.7	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.5	96.4	99.5	102.4	101.9	99.3	97.7	94.8	94.3	95.1	94.4	94.9
		MCWB	76.6	77.4	78.4	80.3	80.9	80.9	81.4	82.0	82.6	81.2	79.8	77.7
	2%	DB	92.5	94.5	97.3	99.8	99.3	96.5	94.8	93.0	92.6	93.2	93.1	92.8
		MCWB	75.9	77.8	79.1	80.5	80.7	80.6	80.0	81.2	81.6	80.4	79.3	77.3
	5%	DB	90.8	93.0	95.4	97.6	97.4	94.5	92.7	91.6	91.2	91.7	91.7	90.8
		MCWB	75.4	77.7	79.1	80.7	80.7	80.7	80.0	80.8	81.3	80.4	78.7	76.7
	10%	DB	88.9	91.3	93.8	95.6	95.2	92.4	91.0	90.1	89.7	90.1	90.0	88.7
		MCWB	74.9	77.3	79.2	80.6	80.9	80.4	79.9	80.3	80.9	80.1	78.0	75.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.0	81.6	84.3	83.9	84.0	83.0	83.0	83.7	85.0	82.8	82.6	80.9
		MCDB	89.4	91.0	93.1	96.1	95.4	93.0	92.8	91.7	92.0	91.3	91.5	90.5
	2%	WB	79.0	80.6	82.1	82.8	82.9	82.1	81.9	82.4	83.0	82.0	81.4	79.7
		MCDB	88.3	90.9	91.6	94.8	93.6	91.8	90.6	90.8	90.1	90.1	90.1	88.9
	5%	WB	78.1	79.7	81.0	81.9	82.0	81.5	81.0	81.6	82.0	81.4	80.4	78.5
		MCDB	86.7	89.5	91.1	93.5	92.1	91.0	89.6	89.6	88.9	89.2	88.6	87.2
	10%	WB	77.0	78.9	80.2	81.2	81.4	80.9	80.3	80.8	81.2	80.7	79.4	77.2
		MCDB	84.7	88.2	90.3	92.3	91.2	90.0	88.6	88.6	87.9	88.2	86.7	85.3
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	17.8	16.7	16.9	15.2	13.8	12.7	12.4	11.9	13.0	15.1	17.3
		MCDBR	20.7	19.1	19.0	18.7	18.1	16.5	15.1	14.1	14.0	15.1	17.0	19.1
		MCWBR	7.4	6.0	5.3	5.1	4.5	4.5	4.6	5.2	5.5	5.4	6.2	6.8
	5% WB	MCDBR	17.5	17.0	17.0	17.3	16.1	14.6	13.7	13.5	13.2	13.9	15.4	17.9
		MCWBR	6.7	5.7	5.7	5.7	5.4	5.0	5.0	5.6	5.8	5.5	6.0	6.8
(40) Clear-Sky Solar Irradiance	taub	0.532	0.617	0.661	0.599	0.511	0.466	0.455	0.457	0.489	0.567	0.509	0.499	(40)
	taud	1.967	1.791	1.724	1.902	2.196	2.324	2.353	2.335	2.234	1.975	2.116	2.082	(41)
	Ebn at Noon	242	227	222	236	253	262	266	268	261	237	247	248	(42)
	Edh at Noon	56	70	77	64	47	41	40	41	45	57	48	49	(43)
(44) All-Sky Solar Radiation	RadAvg	1553	1675	1749	1884	1771	1682	1586	1570	1526	1528	1557	1523	(44)
(45)	RadStd	94	66	117	96	121	90	76	102	79	90	89	62	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (29 neighbors)	+0.40	N/A	N/A	N/A	+0.52	+0.54	N/A	N/A	+184	+184

Nomenclature: See separate page